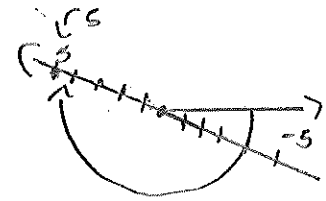
Name [redacted] **ID#** [redacted]

Math 104 Spring 2025 Final Exam

- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

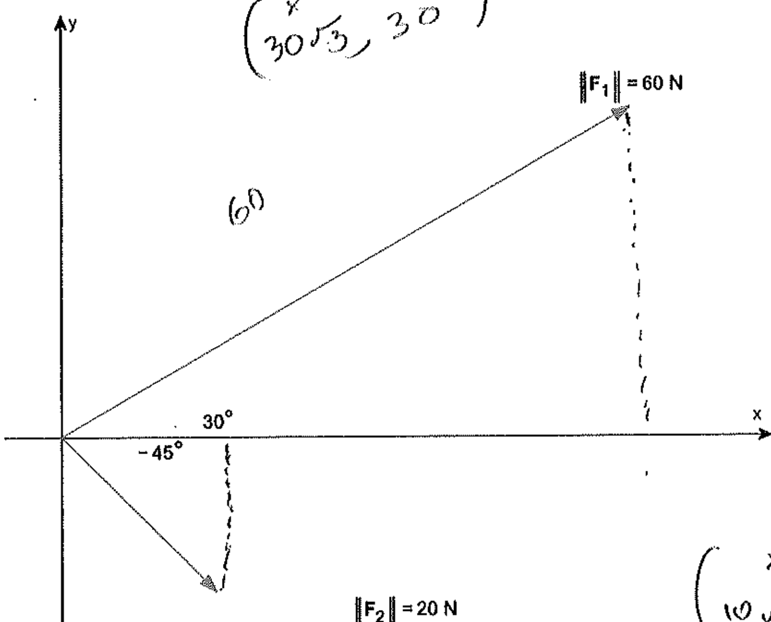
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$

$30° = \pi/6 = \sqrt{3}/2, 1/2 \rightarrow 60 \cdot \frac{\sqrt{3}}{2} = \overset{x}{30\sqrt{3}}, \; 60 \cdot \frac{1}{2} = \overset{Y}{30}$

$(\overset{x}{30\sqrt{3}}, \overset{Y}{30})$



$(\overset{x}{10\sqrt{2}}, \overset{Y}{-10\sqrt{2}})$

$-45° = \frac{7\pi}{4}$

$\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$

(x)

$2\cancel{0} \cdot \frac{\sqrt{2}}{\cancel{2}} = 10\sqrt{2} \qquad \cancel{20} \cdot -\frac{\sqrt{2}}{\cancel{2}} \quad -10\sqrt{2}$

$$\boxed{F_1 + F_2 = (30\sqrt{3} + 10\sqrt{2})i + \left(30 - \frac{20\sqrt{2}}{2}\right)j}$$

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2) Let $g(x) = 4 - x^2$.

$A \quad B$

a) Find the average rate of change from -1 to 3.

$\frac{g(b) - g(a)}{b - a}$

$3, (5)$

$\underline{\underline{B}} = 4 - (3)^2 = \underline{-5}$

$\underline{\underline{A}} \quad 4 - (-1)^2 = 3$

$\frac{-5 - 3}{3 + 1}$

$\frac{-8}{4} \quad \boxed{= -2}$

Avg Rate of change

(slope)

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

Point Slope Formula

$(-1, 3)$

$y - y_1 = m(x - x_1)$

$y + 3 = -2(x + 1)$

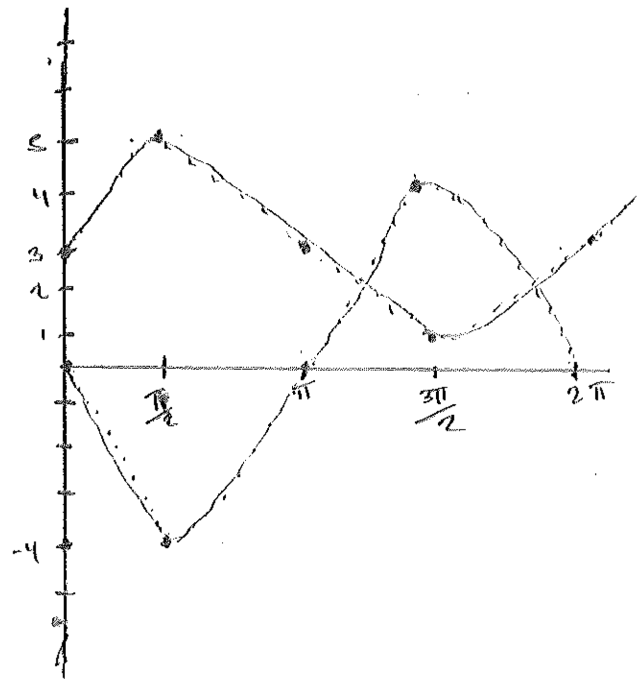
$y + 3 = -2x - 2$

$\quad -3 \qquad\quad -3$

$\boxed{y = -2x - 5}$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin(x) = 2\sin(x) + 3$

$+4\sin x \qquad +4\sin x$

$0 = 6\sin x + 3$

$-3 \qquad -3$

$\frac{-3}{6} = \frac{6\sin x}{6}$

$-\frac{1}{2}$

$\sin(x) = -1/2$

$\boxed{\frac{7\pi}{6} \text{ \& } \frac{11\pi}{6}}$

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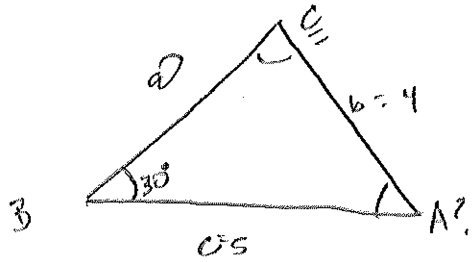
4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$b = 4, c = 5, B = 30^\circ$



$$\boxed{\begin{aligned} C &= 38.7^\circ \\ A &= 111.3^\circ \\ a &= 1.6 \end{aligned}}$$

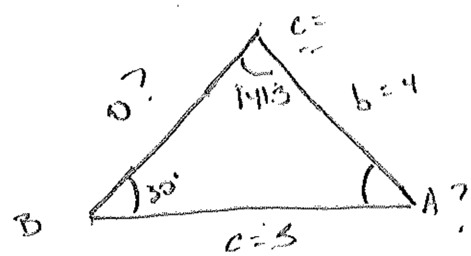
$$\frac{\sin 30^\circ}{4} = \frac{\sin C}{5} = \frac{5 \cdot \sin 30^\circ}{4} = \sin C \rightarrow \sin^{-1}\left(\frac{5 \cdot \sin 30^\circ}{4}\right) \cdot \frac{180}{\pi} = \underline{\underline{38.7^\circ}}$$

$$\frac{\sin 30^\circ}{4} = \frac{\sin 11.3^\circ}{a} = a = \frac{4 \cdot \sin 11.3^\circ}{\sin 30^\circ} = \boxed{1.6} \quad \frac{\sin 38.7^\circ}{5} = \frac{\sin 11.3^\circ}{a} = 1.6$$

2nd triangle check

$180 - 38.7 = 141.3$

$\hookrightarrow 180 - 141.3 - 30 = \underline{8.7^\circ}$ 2nd Triangle Available

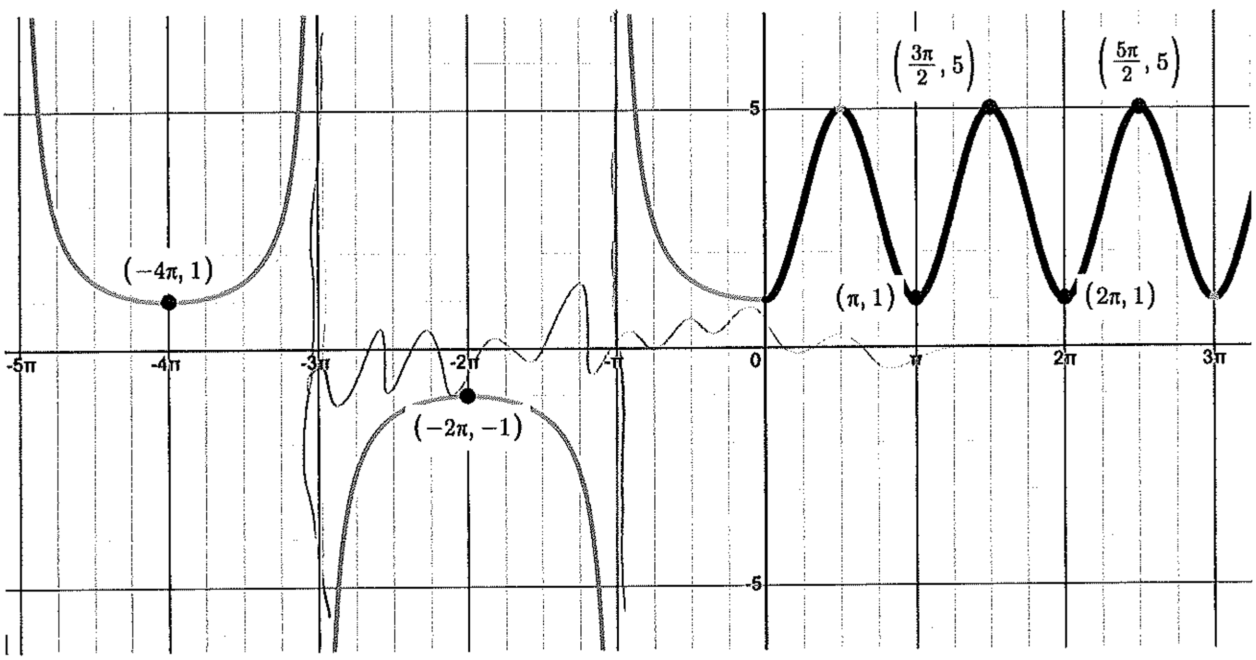


$$\boxed{\begin{aligned} C &= 141.3 \\ A &= 8.7^\circ \\ a &= 1.2 \end{aligned}}$$

$$\frac{\sin 30^\circ}{4} = \frac{\sin 8.7^\circ}{a} = \frac{4 \cdot \sin 8.7^\circ}{\sin 30^\circ} = 1.2$$

$$\frac{\sin 141.3}{5} = \frac{\sin 8.7^\circ}{a} = 1.2$$

5) Given the graph below find the function $f(x)$ and state the domain and range.



$-2\csc\left(-\frac{1}{3}x\right)$

$\sin(2x)+1$

$\frac{\pi}{B}$ $\frac{\pi}{1}$ 3π

$\frac{2\pi}{1}$

$\frac{2\pi}{B} = \pi$

$\frac{2\pi}{\pi}$

$B = 2$

$f(x) = -2\csc\left(\frac{1}{3}x\right)$ when $(-\infty < x \leq 0)$

$F(x) = \sin(2x)+1$ when $(0 \leq x < \infty)$

Range $(-\infty, -1) \cup (1, \infty)$

Domain : all real #'s except $3\pi + k \cup (0 < \infty)$

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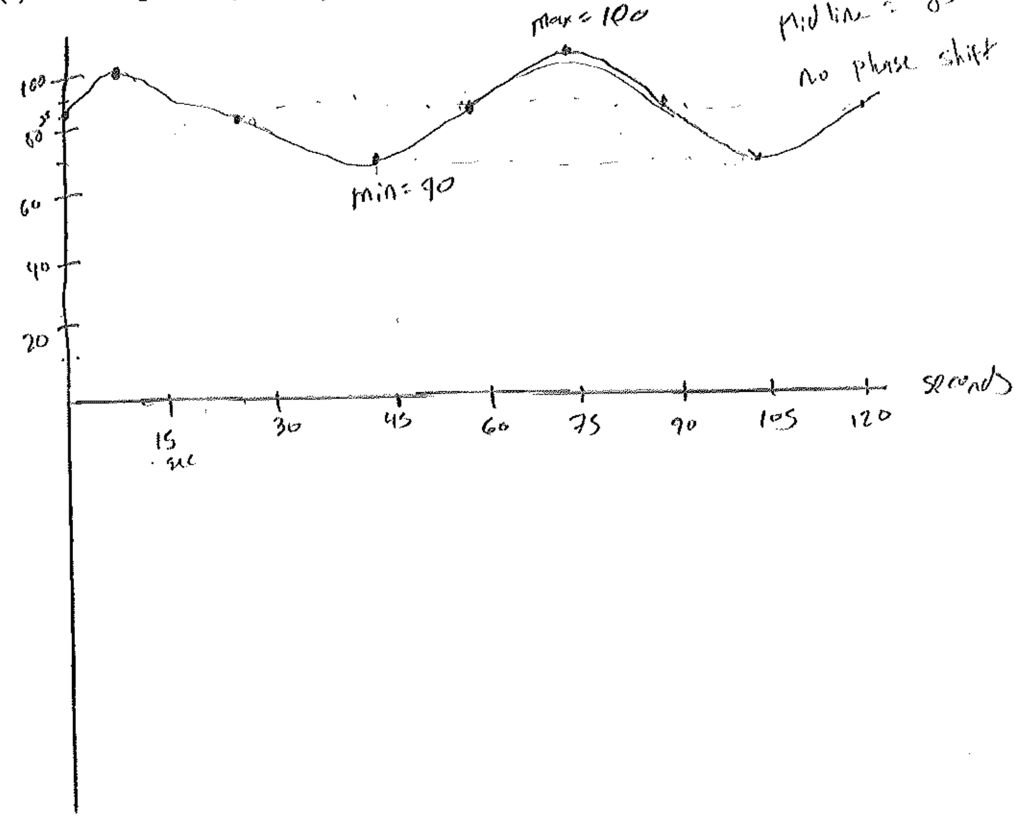
6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

$\frac{2\pi}{2\pi} = 1$ minute sec?

a) Graph $P(t)$ for two periods, label your axes and local maximums.

Amplitude = 15
Mid line = 85
no phase shift



b) The systolic pressure is 70

c) The diastolic pressure is 100

d) The heart rate is 30 beats per minute